

2019 Mathematics Kangaroo Contest

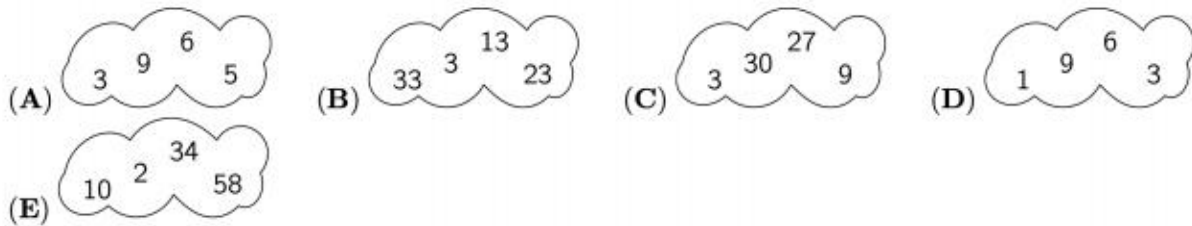
—Cadet—

2019 数学袋鼠竞赛

Part 1: 10 problems, 3 points each | 第一部分: 10 道题目, 每题 3 分

1 • Which cloud contains four even numbers?

哪一朵云上有四个偶数?



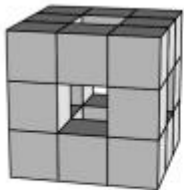
2 • How many hours are there in ten quarters of an hour?

十个四分之一小时是多少小时?

- (A) 40 四十 (B) 5 and a half 五个半 (C) 4 四
(D) 3 三 (E) 2 and a half 两个半

3. A $3 \times 3 \times 3$ cube is built from $1 \times 1 \times 1$ cubes. Then some cubes are removed from front to back, from left to right and from top to bottom, as shown. How many $1 \times 1 \times 1$ cubes are left?

用 $1 \times 1 \times 1$ 的立方体构建了一个 $3 \times 3 \times 3$ 的立方体。然后如图所示, 从前到后, 从左到右, 从上到下移除了一些立方体。问还剩下多少个 $1 \times 1 \times 1$ 立方体?

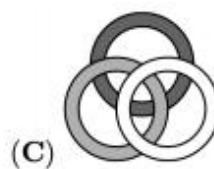
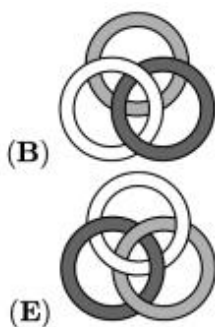
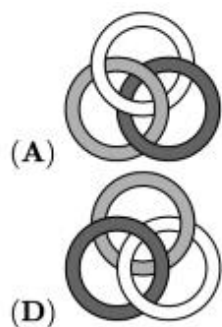


- (A) 15 (B) 18 (C) 20 (D) 21 (E) 22

4. Three rings are linked as shown in the diagram. Which of the following diagrams also shows the three rings linked in the same way?

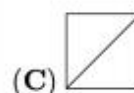
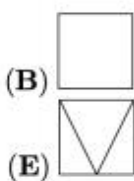
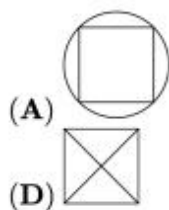
如图所示, 三个环连接在一起。以下哪个图也显示了以相同方式连接的三个环?





5 • Which of the diagrams below cannot be drawn without lifting your pencil off the page and without drawing along the same line twice?

如果要求笔不离纸，并且一条线不能画两次，那么下面哪个图无法画出？



6 • Five friends met. Each of them gave a cupcake to each of the others. They then ate all the cupcakes they had been given. As a result, the total number of cupcakes they had decreased by a half. How many cupcakes did the five friends have at the start?

五个朋友见了面。他们每个人都给了其他每个人一个纸杯蛋糕。然后他们吃了收到的所有纸杯蛋糕。结果，他们手中纸杯蛋糕的总数减少了一半。问五个朋友一开始有多少个纸杯蛋糕？



(A) 20

(B) 24

(C) 30

(D) 40

(E) 60

7. In a race, Lotar finished before Manfred, Victor finished after Jan, Manfred finished before Jan and Eddy finished before Victor. Who finished last of these five runners?

在一场跑步比赛中，Lotar 在 Manfred 之前到达终点，Victor 在 Jan 之后到达终点，Manfred 在 Jan 之前到达终点，并且 Eddy 在 Victor 之前到达终点。问这五名选手中的最后一名是谁？

(A) Victor

(B) Manfred

(C) Lotar

(D) Jan

(E) Eddy

8. The pages of the book Juliet is reading are all numbered. The numbers used on the pages contain the digit 0 exactly five times and the digit 8 exactly six times. What is the number of the final page?

Juliet 正在阅读的书每页都有编号。在页面的编号中，数字 0 恰好出现了五次，数字 8 恰好出现了六次。问最后一页的编号多少？

(A) 48

(B) 58

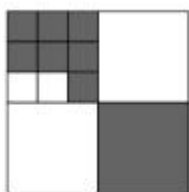
(C) 60

(D) 68

(E) 88

9. A large square is divided into smaller squares. What fraction of the large square is colored grey?

一个大的正方形被分成了一些较小的正方形。灰色的部分占大正方形的几分之几？



(A) $\frac{2}{3}$

(B) $\frac{2}{5}$

(C) $\frac{4}{7}$

(D) $\frac{4}{9}$

(E) $\frac{5}{12}$

10. Andrew divided some apples into six equal piles. Boris divided the same number of apples into five equal piles. Boris noticed that each of his piles contains two more apples than each of Andrew's piles. How many apples does Andrew have?

Andrew 将一些苹果分成相同的六堆。Boris 将同样数量的苹果分成相同的五堆。Boris 注意到他的每一堆都比 Andrew 的每一堆还要多两个苹果。问 Andrew 有多少个苹果？

(A) 60

(B) 65

(C) 70

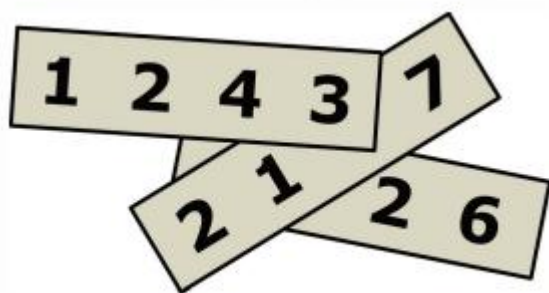
(D) 75

(E) 80

Part 2: 10 problems, 4 points each | 第二部分: 10 道题目, 每题 4 分

11. Four-digit integers are written on each of three pieces of paper. The pieces of paper are arranged so that three of the digits are covered, as shown. The sum of the three four-digit integers is 10126. Which are the covered digits?

在三张纸上分别写有四位整数。如图所示, 这些纸放在一起时有三个数字被覆盖了。三个四位整数的总和是 10126。问被盖住的数字分别是几?



(A) 5, 6, 7

(B) 4, 5, 7

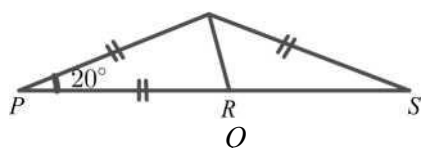
(C) 4, 6, 7

(D) 4, 5, 6

(E) 3, 5, 6

12. In the diagram, $PQ = PR = QS$ and angle $\widehat{QPR} = 20^\circ$. What is the size of angle $\widehat{RQS}$?

如图所示, $PQ = PR = QS$, 并且角 $\widehat{QPR} = 20^\circ$ 。问角 $\widehat{RQS}$ 是多大?



(A) 50°

(B) 60°

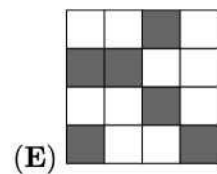
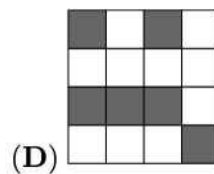
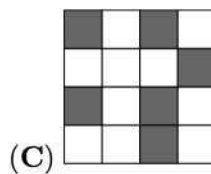
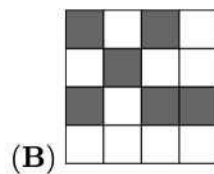
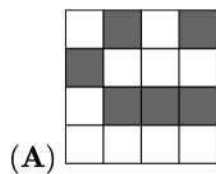
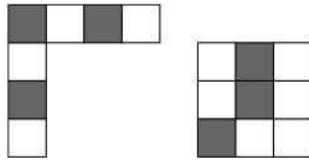
(C) 65°

(D) 70°

(E) 75°

13. Which of the following 4×4 tiles cannot be formed by combining the two given pieces?

哪个 4×4 瓷砖不能够通过下面给定的两片拼成?



14 • Alan, Bella, Claire, Dora, and Erik met at a party and shook hands exactly once with everyone they already knew. Alan shook hands once, Bella shook hands twice, Claire shook hands three times and Dora shook hands four times. How many times did Erik shake hands?

Alan ,Bella,Claire ,Dora 和 Erik 在一个聚会上相遇，并与他们中已经认识的每个人握手一次。Alan 握了一次手，Bella 握了两次手，Claire 握了三次手，Dora 握了四次手。问 Erik 握了几次手？

(A) 1

(B) 2

(C) 3

(D) 4

(E) 0

15 • Jane is playing basketball. After a series of 20 shots, Jane had scored 55% of the time. Five shots later, her scoring rate had increased to 56%. On how many of the last five shots did she score?

Jane 在打篮球。在 20 次投篮后，Jane 的得分率为 55%。又投了 5 次后，她的得分率提高到了 56%。问她的最后五次投篮中有多少次得分？

(A) 1

(B) 2

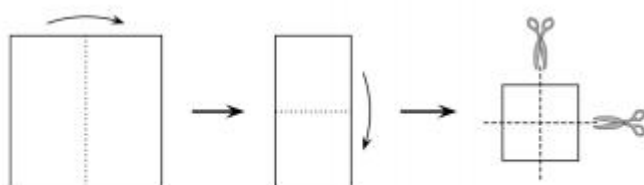
(C) 3

(D) 4

(E) 5

16 • Cathie folded a square sheet of paper exactly in half twice and then cut it in the middle twice, as shown in the diagram. How many of the pieces that she obtains are squares?

如图所示，Cathie 将一张正方形的纸对折两次，然后在中间剪开两次。她会得到多少片正方形？



(A) 3

(B) 4

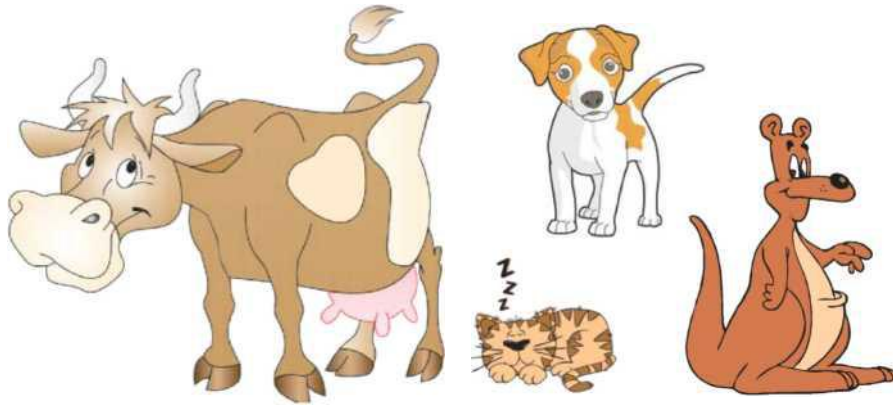
(C) 5

(D) 6

(E) 8

17 • Michael keeps dogs, cows, cats and kangaroos as pets. He tells Helen that he has 24 pets in total and that $\frac{1}{8}$ of them are dogs, $\frac{3}{4}$ are NOT cows and $\frac{2}{3}$ are NOT cats. How many kangaroos does Michael keep?

Michael 将狗、牛、猫和袋鼠作为宠物饲养。他告诉 Helen，他总共有 24 只宠物，其中 $\frac{1}{8}$ 是狗， $\frac{3}{4}$ 不是牛， $\frac{2}{3}$ 不是猫。问 Michael 有多少只袋鼠？



(A)4

(B)5

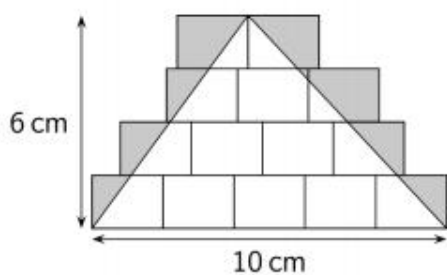
(C)6

(D)7

(E)8

18 • Some identical rectangles are drawn on the floor. A triangle of base 10 cm and height 6 cm is drawn over them, as shown, and the region inside the rectangles and outside the triangle is shaded. What is the area of the shaded region in cm^2 ?

地板上绘制了一些相同的长方形。如图所示，另有一个底边是 10cm，而高是 6cm 的三角形在长方形内，但是在三角形外的区域用阴影标出。问阴影区域的面积是多少 cm^2 ？



(A) 10

(B) 12

(C) 14

(D) 15

(E) 21

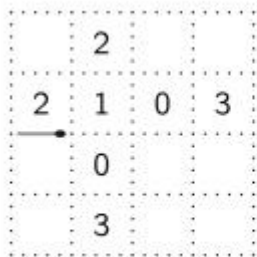
19 • Julio has two cylindrical candles with different heights and diameters. The first candle lasts 6 hours, while the second candle lasts 8 hours. He lit both candles at the same time and three hours later both candles were the same height. What was the ratio of their original heights?

Julio 有两根高度和直径不同的圆柱形蜡烛。第一根蜡烛可燃烧 6 小时，而第二根蜡烛可燃烧 8 小时。他同时点燃两根蜡烛，三小时后两根蜡烛的高度相同。问蜡烛原来的高度比例是多少？

- (A) 4 : 3 (B) 8 : 5 (C) 5 : 4 (D) 3 : 5 (E) 7 : 3

20. Aylin wants to create a path of matches using as few matches as possible. She places each match on the piece of paper like the one shown, along one of the dotted lines. Her path returns to the left-hand end of her original match. The numbers shown in some of the cells are equal to the number of matches around that cell. How many matches are in this path?

Aylin 想要使用尽可能少的火柴来构造一个火柴路径。她将每根火柴放在如图所示的纸的虚线上，其中一根火柴在图中已显示。她的路径的终点就是这根火柴的左端点。某些单元格中显示的数字等于该单元格周围的火柴的数目，问这条路径上有多少根火柴？

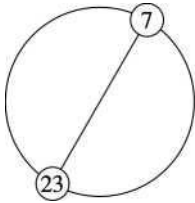


- (A) 12 (B) 14 (C) 16 (D) 18 (E) 20

Part3: 10 problems, 5 points each 第三部分: 10 道题目, 每题 5 分

21 • The integers from 1 to n , inclusive, are equally spaced in order round a circle. The diameter through the position of the integer 7 also goes through the position of 23, as shown. What is the value of n ?

从 1 到 n 的整数(包括首尾两数)按顺序在圆周上等间隔排列。如图所示, 通过整数 7 所在位置的直径也经过 23 所在位置。问 n 是多少?



- (A) 30 (B) 32 (C) 34 (D) 36 (E) 38

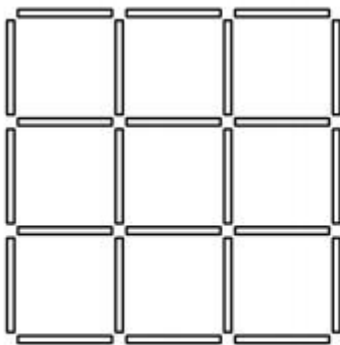
22 • Liam spent all his money buying 50 soda bottles at the store for 1 Euro each. He sells each bottle at the same higher price. After selling 40 bottles, he has 10 Euros more than he started with. He then sells all the remaining bottles. How many Euros does Liam now have?

Liam 花了他所有的钱在商店买了 50 瓶苏打水, 每瓶 1 欧元。他以固定但是更高的价格销售。在卖掉 40 瓶苏打水之后, 他已经比开始时多了 10 欧元。然后他卖掉所有剩下的苏打水。问 Liam 现在有多少欧元?

- (A) 70 (B) 75 (C) 80 (D) 90 (E) 100

23. Natasha has many sticks of length 1. The sticks are coloured either blue, red, yellow or green. She wants to make a 3×3 grid, as shown, so that each 1×1 square in the grid has four sides of different colors. What is the smallest number of green sticks that she could use?

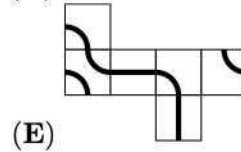
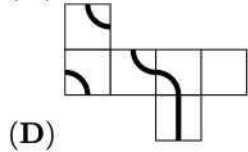
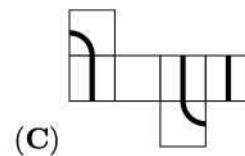
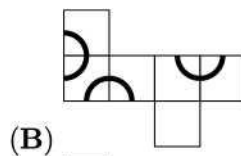
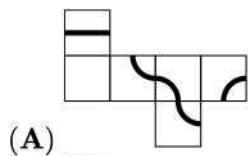
Natasha 有很多长度为 1 的棍子。这些木棒的颜色为蓝色、红色、黄色或绿色。如图所示, 她想要制作一个 3×3 的网格, 使得每个 1×1 正方形的四条边颜色不同。她最少要用多少根绿色的棍子?



- (A) 3 (B) 4 (C) 5 (D) 6 (E) 7

24 • An ant would like to walk along a marked line on the surface of a cube until it returns to its starting point. From which one of the following nets could a cube be made so that such a journey is possible?

一只蚂蚁想要沿着立方体表面上的标记线行走，并最终回到它的出发点。以下哪个展开图做成的立方体，可以使蚂蚁进行这样的行程？



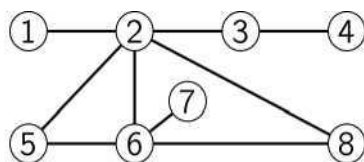
25 • Elisabeta had a large bag of 60 chocolates. She started by eating one 10th of them on Monday, then one 9th of the remainder on Tuesday, then one 8th of the rest on Wednesday, then one 7th on Thursday and so on until she eats half of the remaining chocolates from the previous day. How many chocolates does she have left?

Elisabeta 有一大袋 60 块巧克力。她在星期一吃了十分之一，然后在星期二吃了袋里剩下的九分之一，然后星期三吃了袋里剩下的八分之一，星期四吃了袋里剩下的七分之一，依此类推，直到她吃掉前一天袋里剩下巧克力的一半为止。她还剩下多少块巧克力？

- (A) 1 (B) 2 (C) 3 (D) 4 (E) 6

26 • Prab painted each of the eight circles in the diagram either red, yellow or blue such that no two circles that are joined directly are painted the same colour. Which two circles are necessarily painted the same colour?

Prab 将图中八个圆圈中的每一个都涂上红色、黄色或蓝色的一种，使得没有两个有直线相连的圆圈被涂成相同的颜色。问下面哪两个圆圈必然被涂上相同的颜色？



- (A) 5 and 8 (B) 1 and 6 (C) 2 and 7 (D) 4 and 5 (E) 3 and 6

27 • When Ria and Flora compared their savings, they found that the ratio of their savings was 5 : 3. Then Ria bought a tablet for 160 Euro and the ratio of their savings changed to 3 : 5. How many Euro did Ria have before buying the tablet?

当 Ria 和 Flora 比较他们的存款时，他们发现他们存款的比例为 $5:3$ 。然后 Ria 买了 160 欧元的平板电脑，他们的存款比例变为 $3:5$ 。问在购买平板电脑之前，Ria 有多少欧元？

- (A) 192 (B) 200 (C) 250 (D) 400 (E) 420

28. Some three-player teams enter a chess tournament. Each player in a team plays exactly once against every

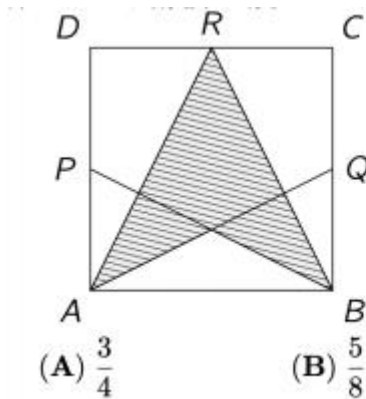
player from all the other teams. For organisational reasons, no more than 250 games can be played in total. At most, how many teams can enter the tournament?

若干三人队参加象棋锦标赛。每个队的每名参赛队员与所有其他队的每个人比赛一次。出于组织原因，比赛的总场次不能超过 250。问最多可以有多少队可以参加锦标赛？

- (A) 11 (B) 10 (C) 9 (D) 8 (E) 7

29 • The diagram shows the square $ABCD$ with P , Q and R the midpoints of the sides DA , BC and CD respectively. What fraction of the square $ABCD$ is shaded?

如图所示，在正方形 $ABCD$ 中， P 、 Q 和 R 分别是边 DA 、 BC 、 CD 的中点。问阴影部分占正方形 $ABCD$ 的几分之几？



- (A) $\frac{3}{4}$ (B) $\frac{5}{8}$ (C) $\frac{1}{2}$ (D) $\frac{7}{16}$ (E) $\frac{3}{8}$

30 • A train is made up of 18 carriages. There are 700 passengers travelling on the train. In any block of five adjacent carriages, there are 199 passengers in total. How many passengers are in the middle two carriages of the train?

一列火车由 18 节车厢组成。火车上共有 700 名乘客。在任何五个相连的车厢里都总共有 199 名乘客。问火车中间的两节车厢里共有多少名乘客？

- (A) 70 (B) 77 (C) 78 (D) 96 (E) 103

答案: EECDD DABDA ABEBC CDBCC BBCEE ACEED